

Part 7

Comparison of Means
Linear Contrasts, Orthogonal Contrasts
Multiple Comparison Procedures

Comparison of Means

- In Part 6, when the hypothesis $H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$ was rejected, the inference is that at least one of the t population means differs from the rest.
- The next question is which means are different from others?
- Is $\mu_1 \neq \mu_2$? Is $\mu_6 \neq \mu_7$?
- Is the average $(\mu_1 + \mu_2 + \mu_3)/3$ different from $(\mu_4 + \mu_5 + \mu_6)/3$? - etc.
- Many times our question will not result in a simple comparison of whether a difference like $\mu_2 - \mu_3 = 0$ or not.

Comparison of Means - cont.

- It may be a more complicated question that requires a comparison like $\mu_1 - (\mu_2 + \mu_3)/2 = 0$ to be made.
- Not all questions can be formulated as comparisons.
- To enable us to understand what kinds of questions can be formulated as comparisons, we define a special linear function of the means.
- A **comparison** among t population means $\mu_1, \mu_2, \dots, \mu_t$ can be written as the linear combination:

$$\ell = a_1\mu_1 + a_2\mu_2 + \dots + a_t\mu_t$$

for given numbers $a_1, a_2, \dots, a_t$ which satisfy $\sum_{i=1}^t a_i = 0$.

- Let us look at some specific examples.

Examples:

Suppose $t = 5$ i.e., we consider the means $\mu_1, \mu_2, \mu_3, \mu_4$, and μ_5 .

- The linear combination $\ell = \mu_2 - \mu_3$ has a_i values

$$a_1 = 0, a_2 = 1, a_3 = -1, a_4 = 0, a_5 = 0.$$

Note that $\sum a_i = 0$ as required for a comparison.

- The linear combination $\ell = (\mu_1 + \mu_2)/2 - (\mu_3 + \mu_4)/2$ has a_i values

$$a_1 = 1/2, a_2 = 1/2, a_3 = -1/2, a_4 = -1/2, a_5 = 0.$$

Again, $\sum a_i = 0$, so it is a comparison.

Linear Contrasts

- A point estimate of a linear combination of population means is called a **linear contrast**, and is given by

$$\hat{\ell} = a_1 \bar{y}_{1.} + a_2 \bar{y}_{2.} + a_3 \bar{y}_{3.} + \cdots + a_t \bar{y}_{t.}$$

with $\sum a_i = 0$.

- The estimated variance of $\hat{\ell}$ is

$$\hat{V}(\hat{\ell}) = s_W^2 \sum_{i=1}^t \frac{a_i^2}{n_i}$$

where n_i is the number of observations taken from the i -th population.

- To test the hypothesis $H_0 : \ell = 0$ we can use the t statistic

$$t = \frac{\hat{\ell}}{\sqrt{\hat{V}(\hat{\ell})}}$$

with degrees of freedom = d.f. for s_W^2 .

Orthogonal Contrasts

- Two contrasts $\hat{\ell}_1 = \sum_i a_i \bar{y}_i$ and $\hat{\ell}_2 = \sum_i b_i \bar{y}_i$ are **orthogonal** whenever $\sum_i a_i b_i = 0$. This is only defined when $n_1 = n_2 = \dots = n_t = n$ i.e., equal sample sizes.
- If all linear contrasts in a set $\hat{\ell}_1, \hat{\ell}_2, \dots, \hat{\ell}_{t-1}$ are pairwise orthogonal, then the set is said to be **mutually orthogonal** set of linear contrasts.
- Given t means $\mu_1, \mu_2, \dots, \mu_t$, and sample means $\bar{y}_1, \bar{y}_2, \dots, \bar{y}_t$ (all based on the same number n of observations), the **maximum number of mutually orthogonal contrasts that exist is $(t - 1)$** .

Orthogonal Contrasts - cont.

- Among t means there are many $(t - 1)$ sets of contrasts that are mutually orthogonal.
- In a maximum mutually orthogonal set

$$\hat{\ell}_1, \hat{\ell}_2, \dots, \hat{\ell}_{t-1},$$

the linear contrasts are random variables which are statistically independent.

- Also, the treatment sum of squares SSB is equal to the sum of the $(\hat{\ell}_i)^2$ for any mutually orthogonal set of $(t - 1)$ contrasts:

$$SSB = \sum_{i=1}^{t-1} (\hat{\ell}_i)^2.$$

SSB has $(t - 1)$ *d.f.* corresponding the $(t - 1)$ contrasts.

Handout - Example 9.3

Consider testing the **Control vs. Agents** comparison:

$$H_0 : \ell = 0 \quad \text{vs.} \quad H_a : \ell \neq 0$$

where $\ell = 4\mu_1 - \mu_2 - \mu_3 - \mu_4 - \mu_5$. The coefficients of the corresponding contrast are therefore:

a_1	a_2	a_3	a_4	a_5
4	-1	-1	-1	-1

Thus

$$\begin{aligned}\hat{\ell} &= 4\bar{y}_1 - \bar{y}_2 - \bar{y}_3 - \bar{y}_4 - \bar{y}_5 \\ &= 4 \times 1.175 - 1.293 - 1.328 - 1.415 - 1.5 = -.836\end{aligned}$$

$$V(\hat{\ell}) = s_W^2 \cdot \sum_{i=1}^5 \frac{a_i^2}{n_i} = (0.153) \left(\frac{20}{6} \right) = .051$$

where

$$\sum_{i=1}^5 \frac{a_i^2}{n_i} = \frac{4^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} = \frac{20}{6}$$

Thus

$$t_c = \frac{\hat{\ell}}{\sqrt{V(\hat{\ell})}} = \frac{-.836}{\sqrt{.051}} = -3.702$$

From Table the t-table, $t_{.025, 25} = 2.06$; thus we reject H_0 at $\alpha = .05$ since $|t_c| > 2.06$ is in the R.R.

We also note that

$$SSC_1 = \frac{(\hat{\ell})^2}{\sum \frac{a_i^2}{n_i}} = \frac{(-.836)^2}{20/6} = .2097$$

and therefore

$$F_c = \frac{SSC_1}{s_W^2} = \frac{.2097}{.0153} = 13.71$$

Since $F_{.05, 1, 25} = 4.24$, we reject H_0 at $\alpha = .05$, the same result as above.

Note carefully that this sum of squares and F-test can be computed instead of the t-test.

Now consider testing the **Biological vs. Chemical** comparison:

$$H_0 : \frac{\mu_2 + \mu_3}{2} = \frac{\mu_4 + \mu_5}{2} \quad \text{vs.} \quad H_a : \frac{\mu_2 + \mu_3}{2} \neq \frac{\mu_4 + \mu_5}{2}$$

Since H_0 is equivalent to $.5\mu_2 + .5\mu_3 - .5\mu_4 - .5\mu_5 = 0$ which is equivalent to $\mu_2 + \mu_3 - \mu_4 - \mu_5 = 0$, the problem is equivalent to testing

$$H_0 : \ell = 0 \quad \text{vs.} \quad H_a : \ell \neq 0$$

where $\ell = \mu_2 + \mu_3 - \mu_4 - \mu_5$. Here the contrast coefficients are

a_1	a_2	a_3	a_4	a_5
0	1	1	-1	-1

giving $\hat{\ell} = 1.293 + 1.328 - 1.415 - 1.5 = -.294$ and

$$V(\hat{\ell}) = s_W^2 \cdot \sum_{i=1}^5 \sum_{i=1}^5 \frac{a_i^2}{n_i} = (0.153) \left(\frac{4}{6} \right) = .0102$$

where

$$\frac{a_i^2}{n_i} = \frac{0^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} + \frac{1^2}{6} = \frac{4}{6}$$

Thus

$$t_c = \frac{\hat{\ell}}{\sqrt{V(\hat{\ell})}} = \frac{-.294}{\sqrt{.0102}} = -2.91$$

Since $t_{.025, 25} = 2.06$; thus we reject H_0 at $\alpha = .05$ since $|t_c| > 2.06$ is in the R.R.

Similar to the previous comparison we may use an F-test:

$$SSC_2 = \frac{\hat{\ell}}{\sum \frac{s_i^2}{n_i}} = \frac{(-.294)^2}{4/6} = .1297$$

and therefore

$$F_c = \frac{SSC_2}{s_W^2} = \frac{.1297}{.0153} = 8.47$$

which leads us to the same result as the t-test as $F_{.05, 1, 25} = 4.24$.

The computations for testing the other two comparisons are similar and are not included here.

Multiple Comparison Procedures

- We know, of course, that the overall error rate when we make multiple tests is larger than α (and possibly much larger).
- To compensate for this, several different multiple comparison procedures have been proposed to control various error rates related to the overall error rate.
- There are several of these; we will consider the following:
 - Fisher's LSD Procedure
 - Tukey's W Procedure
 - Scheffe's Procedure

Multiple Comparison Procedures - cont.

- These are different procedures that each controls a different kind of error rate and each is more or less conservative than others. Each has its set of fans among researchers.
- Each procedure is constructed to control a certain kind of error rate and it is important for a user to be aware of what error rate is controlled by a procedure before using it. We will try to state how conservative each one is as we discuss it.

Fisher's Protected LSD Procedure

The procedure is used for making all possible comparisons between pairs of means $H_0 : \mu_i - \mu_j = 0$ vs. $H_a : \mu_i - \mu_j \neq 0$. It presumes we rejected $H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$.

- For equal sample sizes $n_1 = n_2 = \dots = n_t = n$, consider the t -test of the hypothesis above.

$$t = \frac{(\bar{y}_i. - \bar{y}_j.)}{(s_W \sqrt{2/n})}$$

- We reject H_0 when $|t| \geq t_{\alpha/2}$. This is equivalent to rejecting H_0 , for a pair of (i, j) whenever

$$|\bar{y}_i. - \bar{y}_j.| \geq t_{\alpha/2} s_W \sqrt{2/n}.$$

- The right hand member of this inequality is not a function of i or j . It is constant for a specified α and n , and is called the **Least Significant Difference or LSD**.

- Once the LSD is calculated, doing the tests for the pairs of differences of the form $H_0 : \mu_i - \mu_j = 0$ is simple: Form all possible absolute differences $|\bar{y}_i - \bar{y}_j|$ and reject the corresponding H_0 if this difference exceeds or equals the LSD.
- Testing the hypotheses is thus easy, but **reporting the results** of all those tests can be messy. For t means, there are $t(t - 1)/2$ differences to test.
- To minimize the number of comparisons we need to make, first arrange the $\bar{y}_i$'s **ordered smallest to largest** in value. If we use the notation $\bar{y}_{(i)}$ for the i -th smallest $\bar{y}$, the ranked means may be represented as $\bar{y}_{(1)} \leq \bar{y}_{(2)} \leq \bar{y}_{(3)} \leq \cdots \leq \bar{y}_{(t)}$

- For e.g., $\bar{y}_{(1)}$ might be $\bar{y}_7$ if $\bar{y}_7$ is the smallest; Now note that if the difference $\bar{y}_{(t)} - \bar{y}_{(1)}$, for example, does not exceed the LSD, then all the differences $\bar{y}_{(t)} - \bar{y}_{(2)}, \bar{y}_{(t)} - \bar{y}_{(3)} \dots, \bar{y}_{(t)} - \bar{y}_{(t-1)}$ will not exceed the LSD.
- It follows that in this case we are spared from computing all the above differences and comparing them to the LSD. The following procedure is based on this idea:
- First write the ordered means on a line identified by their corresponding treatment names above them.
- For Example – we might have

trt5	trt3	trt1	trt4	trt2
9.5	10.5	11.6	12.2	13.5

- Take each column in turn, and on a separate line below the list, starting from column 1 connect the means by underlining those pairs of means that are not significantly different from the mean in the current column, in the following way.
- Start the comparison of the mean $\bar{y}_{(1)}$ with the mean on the last column $\bar{y}_{(t)}$. We know that if this pair is less than the LSD value, then none of the differences $|\bar{y}_{(1)} - \bar{y}_{(t-1)}|$ will exceed the LSD value. If so, underline the means connecting $\bar{y}_{(1)}$ with $\bar{y}_{(t)}$
- Otherwise, move left to the next largest mean $\bar{y}_{(t-1)}$ and compare $\bar{y}_{(1)}$ with $\bar{y}_{(t-1)}$, and so on.

- Begin underlining at that column the the difference is found to be less than the LSD value and extend all the way to the left to column 1 (or the column where you started)
- This line implies that those means that are connected with this line are not significantly different from the mean in column 1 and all means between.
- Now restart at column 2 (i.e., $\bar{y}_{(2)}$) and repeat the procedure the same way as above. The new set of underlines will be displayed in a separate line. For Example – we might have

trt5	trt3	trt1	trt4	trt2
9.5	10.5	11.6	12.2	13.5

Example

- Supposed that the computed sample means of six treatments with equal sample size 5 (i.e. $n = 5$) are:
 $\bar{y}_{1.} = 505$, $\bar{y}_{2.} = 528$, $\bar{y}_{3.} = 564$, $\bar{y}_{4.} = 498$, $\bar{y}_{5.} = 600$, $\bar{y}_{6.} = 470$
- Since $MSE = s_W^2 = 2,451$ with 24 d.f. Thus the LSD is:

$$LSD = 2.064\sqrt{2(2451)/5} = 64.63.$$

- Ordered smallest to largest, the means are:

$\bar{y}_{6.}, \bar{y}_{4.}, \bar{y}_{1.}, \bar{y}_{2.}, \bar{y}_{3.}, \bar{y}_{5.} \equiv 470, 498, 505, 528, 564, 600$

- Prepare table to be used in the underlining procedure:

trt6	trt4	trt1	trt2	trt3	trt5
470	498	505	528	564	600

Example - cont.

- Using $LSD = 64.63$, underlining procedure is done as follows:

trt6	trt4	trt1	trt2	trt3	trt5
470	498	505	528	564	600

- Deleting the superfluous lines we have:

trt6	trt4	trt1	trt2	trt3	trt5
470	498	505	528	564	600

Example - cont.

These may lead to one or more of the following conclusions:

- $\mu_6, \mu_4, \mu_1, \mu_2$ are significantly different from μ_5 .
- μ_6, μ_4 are significantly different from μ_3 .
- None of μ_6, μ_4, μ_1 is significantly different from μ_2 .
- None of μ_6, μ_4 is significantly different from μ_1 .
- μ_6 is not significantly different from μ_4 .

When sample sizes are not equal the above procedure is not feasible. In this case, we may construct confidence intervals for all pairs of differences $\mu_i - \mu_j$ using

$$\bar{y}_{i.} - \bar{y}_{j.} \pm t_{\alpha/2} s_W \sqrt{\frac{1}{n_i} + \frac{1}{n_j}}$$

where $t_{\alpha/2}$ is again the percentile from the t-table with degrees of freedom same as that of the within mean square s_W^2 .

Important comments regarding Multiple Comparison procedures

- The protected LSD has a **per-comparison error rate** of α , i.e., the probability of a Type I error is α for any single comparison (or test). However, as we already discussed, the overall error rate when multiple tests are made can be much larger than α , i.e., the probability of making one or more Type I errors exceeds α .
- The **protected** part involves making sure that $H_0 : \mu_1 = \mu_2 = \cdots = \mu_t$ is tested using the analysis of variance F-test prior to using the multiple comparison procedure.

Important comments regarding Multiple Comparison procedures - cont.

- The LSD analysis is carried out only when H_0 is rejected. There is some evidence, based on simulation studies that the **experimentwise error rate** for protected LSD may be near α .
- The **experimentwise error rate** is the probability of observing an experiment with one or more pairwise comparisons falsely declared significant.
- Protected LSD is not a very conservative method. We would not be surprised to see it falsely declare several pairwise comparisons significant in an experiment involving several treatments when all possible differences are tested.

Important comments regarding Multiple Comparison procedures - cont.

- This procedure should not be used to make tests suggested after the experiment has been conducted and the sample means computed. At the planning stage of an experiment, the experimenter must state all questions that needs to be answered in terms of possible comparisons. These comparisons are called **pre-planned or apriori comparisons**.
- Instead of pre-planned comparisons, a part of the plan for the experiment may require testing all differences or only some of them. The intent of LSD is to not to perform all pairwise comparisons routinely.

Important comments regarding Multiple Comparison procedures - cont.

- In any case, it is not recommended that any kind of comparisons be devised after first looking at the $\bar{y}$'s. The problem with testing based on comparisons suggested by looking at the data is that it changes the α level of the test i.e, Type I error rate is not controlled at the specified α level anymore.
- For example — an extreme case — say you look at the sample means and see that the largest is much greater than the smallest, so you decide to test their difference for significance. On average — across experiments — you will seldom fail to reject H_0 when you do this, so the Type I error rate is probably not α .

Tukey's W Procedure

- This method for comparing all possible pairs is **more conservative than LSD** (i.e., it tends to be more resistant to falsely declaring significance.)
- The method is based on comparisons of $|\bar{y}_i - \bar{y}_j|$ to the value

$$W = q_{\alpha}(t, \nu) \sqrt{\frac{s_W^2}{n}}$$

where s_W^2 is the mean square within samples all of size n , ν is the degrees of freedom for s_W^2 , t is number of population means, μ_i compared, and α is the chosen significance level.

Tukey's W Procedure - cont.

- Then if

$$|\bar{y}_i - \bar{y}_j| \geq W$$

we declare that the mean pair μ_i and μ_j are significantly different.

- The value of $q_\alpha(t, \nu)$ is found in Table 10 (Handout). The table gives $q_\alpha(t, \nu)$ for either $\alpha = 0.05$ or $\alpha = 0.01$.
- The the sample means are ordered smallest to largest as before. Then make all possible pairwise comparisons using the value of W and underlining method may then be used to display results.

Example

- The ANOVA table resulting from an experiment involving 6 treatments and $n = 5$ per treatment is:

Source of Variation	DF	SS	MS	F
Between Treatments	5	847.05	169.41	14.37
Within Treatments	24	282.93	11.79	
Total	29	1129.98		

- The sample treatment means are: $\bar{y}_{1.} = 28.8, \bar{y}_{2.} = 24.0, \bar{y}_{3.} = 14.6, \bar{y}_{4.} = 19.9, \bar{y}_{5.} = 13.3, \bar{y}_{6.} = 18.7$
- The ordered means table:

trt5	trt3	trt6	trt4	trt2	trt1
13.3	14.6	18.7	19.9	24.0	28.8

Example - cont.

- From Table 10, $q_{.05}(6, 24) = 4.37$, so

$$W = 4.37 \sqrt{\frac{11.79}{5}} = 6.7$$

- The underlining procedure gives:

trt5	trt3	trt6	trt4	trt2	trt1
13.3	14.6	18.7	19.9	24.0	28.8

Example - cont.

- Means that have an underline in common are declared not significantly different from each other.
- Thus we find that μ_5 , μ_3 , μ_6 , and μ_4 are not different from μ_1 , and μ_5 and μ_3 are not different from μ_2 .
- Just as with LSD, Tukey's method can be used when sample sizes are not all the same, but the above procedure is not feasible. In this case we may construct confidence intervals for all pairs of comparisons $\mu_i - \mu_j$. Its form is

$$\bar{y}_{i.} - \bar{y}_{j.} \pm q_{\alpha}(t, \nu) \sqrt{\frac{s_W^2}{2} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

The value of $q_{\alpha}(t, \nu)$ from Table 10 is obviously the same for all comparisons as well as s_W^2 .

Scheffe's Procedure

- This procedure is ultra conservative. It controls experimentwise error rate. The probability of observing an experiment with one or more contrasts (from the set of all possible contrasts) falsely declared significant is the selected α .
- Scheffe's method can be used to test all possible differences of means (recall that simple differences are contrasts). However it is usually used where contrasts that are not all simple differences are to be tested together with any pairwise differences.
- To test $H_0 : \ell = \sum_i a_i \mu_i = 0$ vs. $H_a : \ell \neq 0$
we base the test statistic on the estimate $\hat{\ell} = \sum a_i \bar{y}_i$.

Scheffe's Procedure - cont.

- Compute the quantity S based on a F -distribution as

$$S = \sqrt{\hat{V}(\hat{\ell})} \sqrt{(t-1)F_{\alpha, df_1, df_2}}$$

- Here $df_1 = t - 1$, and $df_2 = \nu$
- The variance estimate of $\hat{\ell}$ is $\hat{V}(\hat{\ell}) = s_W^2 \sum_i \frac{a_i^2}{n_i}$ where s_W^2 has ν degrees of freedom.
- We reject H_0 when $|\hat{\ell}| > S$.
- Now the underlining procedure is applied in the same way as described for the LSD or Tukey procedures.